A Novel Framework Based on Polarimetric Change Vectors for Unsupervised Multiclass Change Detection in Dual-Pol Intensity SAR Images

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***Abstract*— Change detection (CD) is a crucial topic in many remote sensing applications. In the recent years, satellite polarimetric synthetic aperture radar (PolSAR) systems (e.g., the Sentinel-1 constellation) became a suitable tool for multitemporal monitoring due to the regular acquisitions with a short revisit time in different polarimetric channels. Methods for CD in PolSAR data mainly focus on binary CD (i.e., they provide information about the presence/absence of change only), whereas the polarimetric enhanced information provides multiple features that can be exploited for performing multiclass CD. In this article, we introduce a novel framework for the characterization of multitemporal changes in dual-polarimetric data. The framework is based on the definition of polarimetric change vectors (PCVs) and their representation in a polar coordinate system. PCVs allow characterizing and, thus, to separate multiclass changes in terms of target properties of the single-time scenes and the scattering theory. The proposed model is used to: 1) derive the statistical behaviors of change and no change classes in PolSAR multitemporal images; 2) design an automatic and unsupervised strategy to estimate the optimal number of changes; and 3) distinguish no change from change classes and the kinds of change from each other. An experimental analysis has been conducted on three multitemporal PolSAR data sets having different complexities in terms of number and kinds of change classes. The results confirm the effectiveness of the proposed approach and the better performance with respect to both specific techniques for CD in dual-pol SAR data and a general multiclass CD method, not designed for PolSAR data.**

***Index Terms*— Change detection (CD), expectation maximization (EM), multiclass unsupervised CD, polarimetric change vector (PCV), polarimetric synthetic aperture radar (PolSAR).**

I. Introduction

THE temporal monitoring of the Earth is a crucial topic for several applications. Some of them consider the analysis of long-time series for studying the impact of climatic events at large time scale [1]. Others focus on the analysis of the impact of abrupt changes (e.g., in the case of natural disasters [2]–[4]) and are based on a steady monitoring of a given study area. In this context, change detection (CD) is defined as the process of identifying changes by analyzing images acquired at different times over the same geographical area. Spaceborne synthetic aperture radar (SAR) images represent a powerful tool for CD, as their acquisition is not affected by the cloud coverage and the sun illumination conditions. These properties increase the amount of regular acquisitions that can be used on a given scene compared to passive sensors, especially in extremely cloudy areas.

SAR systems provide the measures of backscattering properties of a scene by transmitting and receiving with linear polarizations (i.e., H or V) [5]. The interpretation of the information measures is a complex task, as SAR imagery is characterized by both geometrical distortions over nonflat targets and multiplicative speckle noise.

Because of the properties of the SAR data, target discrimination in either single-time or multitemporal analysis requires appropriate methodologies. In the literature, several CD approaches based on SAR data have been presented. Few of them are supervised and perform the comparison of multitemporal classification maps generated from single-time SAR images (e.g., postclassification comparison methods [6]). Most of the others are based on the comparison of multitemporal SAR images and on unsupervised information extraction from the resulting change index (CI). The CI can be derived either by a pixel-by-pixel basis or by exploiting the spatial context information. The latter option mitigates the effects of the speckle noise on the results. Several comparison operators have been proposed for the generation of the CI that can be selected on the basis of the kind of considered data and their statistics. In SAR data, in areas with uniform texture, pixel intensity is often modeled with a Gamma probability density function, and the noise is modeled as being multiplicative. These elements support the use of ratio-based comparison operators [7], including normalized-mean ratio [8], [9], log-ratio [5], [10]–[12], or Gauss log-ratio [13]. The representation in the logarithmic scale has the double effect of transforming the residual noise into an additive contribution and enhancing ratio values smaller than 1, which corresponds to backscattering decrease. Other approaches are based on distance measures (e.g., Kullback–Leibler distance [14]) or features coming

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from the information theory (e.g., mutual information [15]) and take the advantage of spatial neighborhood information.

The abovementioned approaches rely on the use of SAR imagery with a single polarimetric channel and detect the presence/absence of changes with the only possible distinction between the positive and/or negative variation of the scattering intensity. If more classes of change on the ground contribute to intensity variations, they fall into either the positive or the negative backscattering variation without further distinctions. In other words, the backscattering variation on a single polarimetric channel provides only limited information about the nature of the kind of change, so that, a multiclass CD is often not feasible or limited to the detection of backscattering increase and decrease, only. In this context, polarimetric SAR (PolSAR) sensors extend the SAR imaging capabilities by transmitting and/or receiving on more polarization channels [16], thus providing a more powerful tool for multiclass CD. Two types of polarimetric data are available, namely, full-polarimetric (full-pol) and dual-polarimetric (dual-pol). In full-pol data, all the combinations are exploited by transmitting and receiving on the two alternate polarizations (H and V). In dual-pol data, only two of the combinations are taken. In some cases, the two combinations are related to a fixed transmitting polarization (i.e., HH–HV or VV–VH). In others, only co-polarizations (i.e., HH–VV) are considered. In this article, we focus on the use of dual-pol intensity data with a fixed transmitting polarization.

This is a very important case given the properties of many satellite systems, including the Sentinel-1 constellation, which is presently one of the major sources of multitemporal data. CD applications exploiting polarimetric data are typically based on likelihood ratio tests [16], [17], the fusion of change maps obtained by single polarimetric features [16], or the descriptive analysis of the features coming from the polarimetric decompositions [3], [5], [10], [16], [18]. However, to the best of the authors' knowledge, the aforementioned CD approaches aim at performing binary CD (i.e., they identify the presence/absence of change without considering that changes may have different causes and, thus, semantic). Sometimes they separate among increase and decrease of the backscattering, but none of them distinguish among several possible multitemporal backscattering mechanisms associated with different kinds of change on the ground [3]. Nevertheless, the use of the polarimetric information in a proper multitemporal framework opens to the possibility of performing multiclass CD since different changes on the ground induce different scattering intensity variations on the polarimetric channels. An effective extraction of such information may lead to a CD map with a larger semantic content. This aspect is crucial in applications that require the identification of the sources of change. For example, in emergency management situations after natural disasters, where changes in urban areas should be detected and analyzed, the possibility of separating different kinds of change, rather than having a single macrochange class without distinction, often significantly improves the effectiveness of rescue activity organization. In multitemporal multispectral images, unsupervised multiclass CD has been tackled by the definition of both multidimensional CIs and proper techniques for the information visualization and extraction [19], [20]. However, those methods cannot be applied to PolSAR data for both theoretical and practical reasons [21]. Thus, novel design and formulation are required for tackling the multiclass CD in PolSAR imagery. A first attempt in this direction was proposed by Pirrone *et al.* [4], introducing an early concept of change vector for fully polarimetric data. However, the concept was limited to fully polarimetric data and was not supported by both a theoretical analysis and techniques for an effective use of polarimetric information in the design of a fully automatic multiclass CD.

In this article, we formulate a novel, complete and detailed theoretical framework for the definition of the multiclass CD problem in bitemporal dual-pol SAR images. This framework is different from the previous multiclass CD approaches [19], [22]. The framework is supported by a theoretical analysis of both the PolSAR signal behaviors in both single- and multidate images and the statistical distribution of the proposed polarimetric change vector (PCV) in polar coordinates. The framework includes a novel theoretical/conceptual model for the interpretation of the physical meaning of changes and the corresponding decision regions in the PCV domain. This is crucial to understand the semantic of changes. The framework also includes a technique for the definition of the decision boundaries between classes, which is based on a derived statistical modeling of the distribution of change and no-change classes along magnitude and direction variables. In particular, a novel Nakagami mixture model for the statistical distribution of the PCV magnitude is introduced, and an adaptation of the expectation maximization (EM) algorithm for the automatic estimation of the unknown mixture parameters is derived [4]. It is worth noting that the strategy is proposed for medium-/high-resolution SAR images, so that, samples can be assumed as independent and identically distributed. The framework is not valid for very-high-resolution SAR images.

This article is structured into five sections. Section II provides an overview of the single-date polarimetric scattering sensitivity with respect to the main target parameters. Section III defines the theoretical multitemporal model for dual-Pol SAR imagery, introduces the PCV, and illustrates a set of possible kinds of change that can be encountered in PolSAR data and their multitemporal behaviors in the proposed feature space (with examples). Section IV presents the proposed automatic decision strategy for multiclass CD in dual-pol SAR data. Section V illustrates the theoretical behaviors of changes in the proposed PCV feature space and presents the experimental results for three data sets and a total of seven different kinds of multiclass changes. Finally, Section VI draws the conclusions and illustrates future developments.

II. Overview of the Scattering Sensitivity Analysis for Target Properties

This article deals with the multiclass CD in bitemporal PolSAR images, acquired at times t_1 and t_2. The expected output is a map showing a no-change class ω_{nc} and K classes of change $\omega_{ck}, k \leq K$, where k specifies that there are different classes of change in terms of scattering properties and, thus, semantic meaning. Here, a conceptual/theoretical sensitivity

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analysis of the polarimetric scattering with respect to the main sensor/target parameters is conducted at single date. The outcome of this analysis is then used for modeling the expected multitemporal scattering behaviors and, thus, the changes in the PCV domain.

The backscattering sensitivity analysis is conducted with respect to the parameters of both the sensor and the target. Among them, we examine the polarimetry information and the frequency band as sensor parameters. However, an extended backscattering analysis is out of the purpose of this article, and further details can be found in [23] and [24].

Polarimetric channels show a different sensitivity to targets with different structures based on the scattering theory. Among the co-pol channels, the VV channel is more sensitive to the horizontal elements, while the HH one is more sensitive to the vertical structures with dihedral elements. The cross-polarimetric channel VH highlights the presence of depolarizing elements [25].

The selection of the frequency band of the SAR system highlights different scene layers, as it directly affects the capability of penetration (together with the dielectric constant) and the sensitivity to roughness of the incident wave [24], [26]. Systems with lower frequencies (i.e., P- and L-band systems) are able to penetrate and detect targets on deeper layers with high co-pol scattering, while at higher frequencies (i.e., C- and X-band systems), they sense information on the shallower layers [27]–[29]. The information of the deeper layers may be seen as cross-polarized signal because of the depolarizing effect of the target internal structure.

Sensor parameters can be assumed to be stable over acquisitions, such that backscattering variations can be defined in terms of the target parameters only. They include the dielectric constant, the surface roughness, the local incidence angle, the aspect angle, and the presence of dihedral elements.

The dielectric constant, associated with the bio-physical properties of the target, affects both the penetration depth and the backscattering intensity. One of the elements impacting on the dielectric constant is the water content. Several studies illustrated how high values of moisture content and dielectric constant are associated with large backscattering values on all the polarimetric channels [30], [31].

Surface roughness is modeled by root-mean-square height s (or facets slope statistical parameter β) and one or two scales, depending on its complexity [30], [32]. For smooth surfaces (i.e., relatively low s values), scattering models show low values for both co- and cross-pol values (i.e., specular reflection). In particular, scattering intensities on the two co-pol channels show similar small values, while the cross-pol intensity tends to zero. Rough surfaces (i.e., relatively large s values) with a single roughness scale, such as bare soils, show scattering based on a general Bragg model with both co-pol and cross-pol nonnull values [32]. Sea surface is typically characterized by two roughness scales, associated with capillary and gravity waves [33], [34]. Capillary waves have very small roughness scale. Accordingly, scattering (under Kirchoff approximation) presents low cross-pol values and sensible co-pol values. Gravity waves modulate the slope scale of the capillary waves and increase the scattering on the co-pol channels only. An additional non-Bragg term on both the co-pol channels is introduced for very rough marine surfaces.

The target signal is also characterized by the aspect angle and the local incidence angle. The aspect angle plays an important role in the analysis of dihedral elements (e.g., built-up areas [25]). These targets generally show high scattering on co-pol channels, while cross-pol scattering depends on the target aspect angle and it has a minimum value for angles close to zero.

The local incidence angle has an impact on the measure of the target geometry in terms of both cell resolution and geometrical distortions for nonflat targets. In particular, large shadow and foreshortening effects are associated with large- and small-angle values, respectively. From a radiometric perspective, under the assumption of lambertian surface, large values are associated with lower backscattering intensities [26].

The presence of dihedral elements in the resolution cell introduces a deterministic scattering contribution, which depends on the target aspect angle. Such a contribution generates high backscattering values on all the polarimetric channels (depending on the aspect angle) and mitigates the effect of the sum of the contributions from the other scattering elements in the resolution cell. This is exploited for the detection of urban areas in the SAR imagery [3], [35].

The aforementioned date backscattering analysis holds for each SAR image whatever the acquisition time is. In Section III, changes will be modeled as multitemporal combination of the mentioned single-date backscattering behaviors.

III. Proposed Intensity-Based PCV Analysis

Let us consider two intensity dual-pol SAR images X_1 and X_2 with size $M \times N$, acquired over the same scene at times t_1 and t_2 ($t_1 < t_2$), respectively. Let us assume that preprocessing step is applied to the pair of images (i.e., calibration, multilooking, geocoding, and co-registration). By considering intensity data, off-diagonal elements of the matrix C are neglected; thus no correlation information is assumed between the two polarimetric channels. Each polarimetric image X_i ($i = 1, 2$) is modeled as a data-cube structure, in which the generic spatial location (x, y) is associated with a vector $X_i(x, y)(x = 1, \ldots, M; y = 1, \ldots, N)$ with two components. Let p be the fixed transmission polarization and q its orthogonal polarization. X_i is defined as follows:

$$X_i(x, y) = \left[X_i^{pp}(x, y), X_i^{pq}(x, y)\right], \quad p, q \in \{\mathrm{H}, \mathrm{V}\}, \ \mathrm{p} \neq \mathrm{q} \tag{1}$$

where $X_i^{pp}(x, y)$ and $X_i^{pq}(x, y)$ are the backscattering coefficients measured for the pixel in position (x, y) of the ith image on the co- and cross-polarized channels, respectively.

A. PCV

Starting from the single-date backscattering behavior in X_1 and X_2 (see Section II), we perform a theoretical/conceptual analysis of the physical and semantic meaning of changes. This is done by proposing a novel feature space based on PCVs

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to effectively represent and interpret change information. The proposed model is based on a polarimetric multitemporal comparison of the input images and on the subsequent representation of the polarimetric multitemporal information in the dual-pol log-ratio feature space. The comparison operator is a polarimetric version of the log ratio

$$X_{\mathrm{LR}} = \left[X_{\mathrm{LR}}^{pp}, X_{\mathrm{LR}}^{pq}\right] \tag{2}$$

$$X_{\mathrm{LR}}^{pp} = \log \frac{X_2^{pp}}{X_1^{pp}} \tag{3}$$

$$X_{\mathrm{LR}}^{pq} = \log \frac{X_2^{pq}}{X_1^{pq}}. \tag{4}$$

The output X_{LR} is named as PCV. Each PCV component refers to a polarimetric channel and, thus, codes different characteristics of the change information. Log-ratio components in the PCV have the positive/negative values associated with the increase/decrease of the SAR intensity in the corresponding polarimetric channel and, thus, multiclass changes. The PCVs span a 2-D orthonormal feature space. The information in this feature space can be represented in a two 2-D coordinate systems: the Cartesian one (given by the log-ratio values for the two polarimetric channels) or the polar one. The relationship between the polar and Cartesian coordinates is given by

$$\rho = \sqrt{\left(X_{\mathrm{LR}}^{pp}\right)^2 + \left(X_{\mathrm{LR}}^{pq}\right)^2}, \quad \rho \geq 0 \tag{5}$$

$$\phi = \arctan \frac{X_{\mathrm{LR}}^{pq}}{X_{\mathrm{LR}}^{pp}}, \quad \phi \in [0, 2\pi] \tag{6}$$

where ρ is the magnitude and ϕ is the direction of the PCV. No-change and change classes can be characterized according to the magnitude and direction properties of the PCVs. On the one hand, the no-change class does not exhibit any relevant backscattering variation in time both in the co- and cross-pol channels. Both the PCV components assume values close to 0. Thus, ρ is expected to have values close to 0 as well. On the other hand, change classes are associated with significant backscattering time variations (either positive or negative) in one or both the polarimetric channels (i.e., at least one of the conditions $X_1^{pp} \neq X_2^{pp}$ and $X_1^{pq} \neq X_2^{pq}$ is satisfied). Therefore, at least one of the two PCV elements is significantly different from zero. In these cases, magnitude values are in average higher than those of the no-change class.

Let ω_{nc} be the class of unchanged pixels. The complementary class Ω_c represents the macroclass of all the possible kinds of change. In the feature space, classes ω_{nc} and Ω_c can be separated with a magnitude threshold T_ρ. This results in a circle C_{nc} (circle of unchanged pixels) and an annulus A_c (annulus of changed pixels), which are defined as follows:

$$C_{nc} = \{\rho, \phi : 0 \leq \rho \leq T_\rho, 0 \leq \phi < 2\pi\} \tag{7}$$

$$A_c = \{\rho, \phi : \rho > T_\rho, 0 \leq \phi < 2\pi\}. \tag{8}$$

For the no-change class, the direction values are almost uniformly distributed and, thus, exist in the whole domain of ϕ. Within C_{nc}, therefore, the direction provides no relevant information for the no-change class. On the contrary, the direction is important for discriminating among the change classes in the macroclass of changes Ω_c, as it assumes different values within A_c according to the effects of changes on the physical parameters of the scene. Different change classes assume positive, null, or negative values for each PCV component. They correspond to different possible direction values in the PCV polar domain. For the sake of simplicity, we assume eight possible preferred direction values. Given the polar representation, the first four directions $\phi \in \{\pi/4, 3\pi/4, 5\pi/47\pi/4\}$ correspond to the cases in which the absolute values of the two PCV elements X_{LR}^{pp} and X_{LR}^{pq} are similar to each other. The second set of directions $\phi \in \{0, \pi/2, \pi, 3\pi/2\}$ includes the cases in which the absolute value of one component is much stronger than that of the other. Each kind of change can be mapped into a region of the PCV feature space surrounding its specific preferred direction. Let $\omega_{ck}, k = 1, \ldots, 8$ be the kth change class, where

$$\Omega_c = \bigcup_{k=1}^{8} \omega_{ck}. \tag{9}$$

Each class ω_{ck} is mapped into an annular sector S_k having magnitude greater than T_ρ and limited by the thresholds T_{k1}^ϕ, T_{k2}^ϕ along the direction. The first sector presents a discontinuity for $\phi = 0$. This discontinuity makes the sector S_1 as the union of two intervals defined by thresholds T_{11}, T_{12}

$$S_k = \left\{\rho, \phi : \rho > T_\rho, T_{k1}^\phi \leq \phi < T_{k2}^\phi, T_{k1}^\phi \leq T_{k2}^\phi\right\}, \quad k \neq 1 \tag{10}$$

$$S_1 = \left\{\rho, \phi : \rho > T_\rho, 0 \leq \phi < T_{12}^\phi, T_{11}^\phi < \phi < 2\pi, T_{11}^\phi \geq T_{12}^\phi\right\}. \tag{11}$$

Despite the sensitivity of the backscattering to a large set of physical variables, three main variables are individuated that characterize land-cover backscattering, namely, the local roughness, the water content, and the presence of targets with nonzero height. Such parameters have been used for the analysis of the backscattering behavior in X_1 and X_2 (see Section II). Changes on the ground induce variations on those physical parameters that will result in variations of the backscattering values from X_1 to X_2 and, thus, in different behaviors of X_{LR}. Therefore, they are used in Section III-B to derive different behaviors of multiclass changes.

B. Target Change Modeling in the PCV Domain

Here, we extend the single-time scattering analysis in Section II to a multitemporal perspective and illustrate the multiclass changes that can be identified by means of dual-pol SAR images and characterize their behaviors in the proposed PCV domain.

If characteristics in terms of physical parameters are preserved (i.e., no change occurred), small backscattering variations for both the polarimetric channels are mainly attributable to the noise effects (i.e., $X_{\mathrm{LR}}^{pp}, X_{\mathrm{LR}}^{pq} \simeq 0$). On the other hand, if there is a change in one or more physical properties, relevant backscattering variations are detected on one or both the polarimetric channels based on the analysis illustrated in Section III-A. Physical changes generating similar backscattering variations in the two polarimetric channels due to dual-pol data limitations are considered as being the same change class.

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In the following, variations in the target physical properties are analyzed and associated with backscattering behaviors in the PCV domain.

1) Changes in the Dielectric Constant: Changes in the target dielectric constant can be due to different reasons, but are typically associated with the variation of the target water content. A positive variation is associated with a direction $\phi = \pi/4$ corresponding to a backscattering increase on both the polarimetric channels (i.e., $X_{\mathrm{LR}}^{pp}, X_{\mathrm{LR}}^{qq} > 0$); conversely, a decrease of the water content is associated with a decrease of backscattering value on both the channels with a corresponding direction $\phi = 5\pi/4$. The positive variations of the water content are seen after rainfall events or water content variations like the ones due to irrigation. The negative variations happen in the presence of seasonal aging and the loss of biomass of the vegetation [36]. Snow is another example of target that may show significant changes in the dielectric constant due to water content. However, the effects on the backscattering are more complex. The increase of the water content in snow is associated with both an increase of the dielectric constant and a variation of the layer penetration depth. Thus, the wetness increase of the snow is associated with a backscattering decrease on both the polarimetric channels (i.e., $X_{\mathrm{LR}}^{pp}, X_{\mathrm{LR}}^{pq} < 0$) [37], [38].

2) Change in the Surface Roughness: Let us consider variations in the surface roughness (e.g., a bare soil or a water body). In the case of soils, where roughness is at a centimeter scale (i.e., much smaller than the image geometrical resolution), an increase of the roughness parameter is seen as backscattering increase on both the co-pol and cross-pol channels (i.e., $X_{\mathrm{LR}}^{pp}, X_{\mathrm{LR}}^{pq} > 0$). This change is associated with the preferred direction $\phi = \pi/4$ and may occur in the presence of soil tillage [23]. The opposite variations of the soil roughness show the negative scattering variations (i.e., $X_{\mathrm{LR}}^{pp}, X_{\mathrm{LR}}^{pq} < 0$) and are mapped on the direction $\phi = 5\pi/4$. On the other hand, water surface is composed of capillary waves with very small roughness scale modulated by waves with larger scales. This variation affects scattering mainly on the co-pol channel with negligible effects on the cross-pol one (i.e., $X_{\mathrm{LR}}^{pq} \simeq 0$). The positive and negative roughness variations are mapped into the corresponding directions $\phi = 0$ (i.e., $X_{\mathrm{LR}}^{pp} > 0$) and $\phi = \pi$ (i.e., $X_{\mathrm{LR}}^{pp} < 0$) of the polar domain, respectively. These changes may occur in the presence of a variation of the sea swell or an oil spill, which univocally reduces the surface roughness [39].

3) Changes in the Dihedral Elements: They are associated with the change of the dihedral structures in the resolution cell and show a backscattering variation for both the polarimetric channels with preferred directions either $\phi = \pi/4$ or $\phi = 5\pi/4$ for a dihedral target appearing or disappearing, respectively. The variation on the cross-pol channel is smaller if the aspect angle decreases and becomes very small in the presence of targets with negligible aspect angle. As $X_{\mathrm{LR}}^{pq} \simeq 0$, the preferred direction for the two changes tends to $\phi = 0$ and $\phi = \pi$, respectively. These changes may occur when large targets appear on the sea (e.g., ships) [40] or when there is a growth of large vegetation [41]. The variation of multiple target parameters may occur for complex changes and depends on collateral conditions.

4) Change in Multiple Target Parameters—Example of Floods: A flood event may occur on either barely or largely vegetated soil. In the case of the barely vegetated soil, under the assumption of smooth water surface, the effect on the backscattering by the increase of the soil dielectric constant is overtaken by the decrease of the surface roughness. This generates a backscattering decrease on both the polarimetric channels (i.e., $X_{\mathrm{LR}}^{pp}, X_{\mathrm{LR}}^{pq} > 0$) with corresponding preferred direction $\phi = 5\pi/4$. On the other hand, in the case of largely vegetated soil, the flooded scene presents some dihedral structures formed by the trees and the water surface. This results in no relevant backscattering variations on the co-polarimetric channel between the two acquisitions (i.e., $X_{\mathrm{LR}}^{pp} = 0$) and a backscattering decrease on the cross-pol channels, respectively. The corresponding preferred direction resulting from the two backscattering variations results to be $\phi = \frac{3\pi}{2}$. When the assumption of smooth water surface is not fulfilled (i.e., rough water surface), the backscattering variation on the co-pol channel takes into account a further backscattering increase associated with the increase of the surface roughness with respect to the case of smooth water surface.

5) Change in Multiple Target Parameters—Example of Changes in Built-Up Areas: The second example of complex changes is associated with the presence of built-up areas, which are characterized by dihedral structures with large backscattering. In particular, the construction of built-up areas on a bare soil is characterized by a backscattering increase on both the polarimetric channels (i.e., $X_{\mathrm{LR}}^{pp}, X_{\mathrm{LR}}^{pq} > 0$), due to the appearance of the dihedral structures. The two backscattering variations result in a preferred direction $\phi = \pi/4$. On the other hand, the construction on areas with either vegetation or debris (debris might be generated by collapsed urban areas, for example) is characterized by the same variation on the co-pol channel (i.e., $X_{\mathrm{LR}}^{pp} > 0$) and a stable backscattering on the cross-pol channel (i.e., $X_{\mathrm{LR}}^{pq} = 0$), with a preferred direction equal to $\phi = 0$. Conversely, the destruction of built-up areas (e.g., in the presence of seismic events) is associated with a decrease of the scattering in the co-pol channel (i.e., $X_{\mathrm{LR}}^{pq} < 0$). A relevant decrease on the cross-pol channel is also present when no debris exists on the postevent scene (i.e., $X_{\mathrm{LR}}^{pq} < 0$). The behavior can be explained by the high values of cross-pol scattering associated with the vegetation, the debris, and the oriented structure [2], [3], [42].

6) Change in Multiple Target Parameters—Example of Changes of Vegetated Areas: The last example is represented by deforestation phenomena (e.g., wildfire or clear-cutting on dense vegetated areas). The effect of such phenomena is strongly dependent on the frequency band. For higher frequencies (i.e., C- or X-band), the radiation has the capability to penetrate the vegetation canopy. The removal of the canopy keeps the trunk vertical structures, which form dihedral elements with the ground. Such changes result in backscattering increase and decrease on the co-pol and cross-pol channels, respectively (i.e., $X_{\mathrm{LR}}^{pp} > 0$, $X_{\mathrm{LR}}^{pq} < 0$). The change is mapped into a preferred direction $\phi = 7\pi/4$. If the trunk structure is cleared

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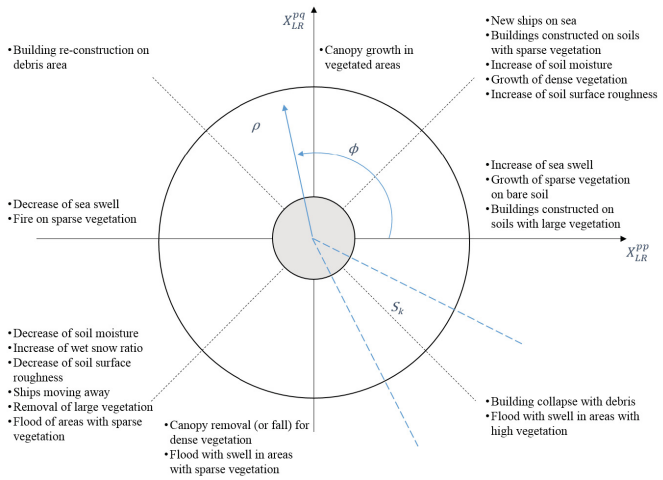


Fig. 1. Change behavior in the polar PCV feature space.

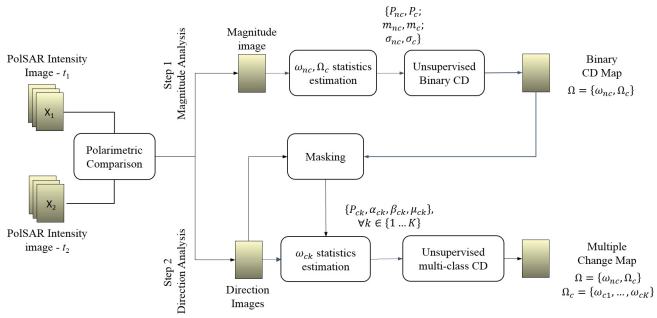


Fig. 2. Block diagram of the CD method.

as well, the scattering is sensed from the bare soil below the vegetation. Thus, the co-pol scattering shows no relevant backscattering variation (i.e., $X_{\mathrm{LR}}^{pp} \simeq 0$), while the co-pol scattering decreases (i.e., $X_{\mathrm{LR}}^{pq} < 0$) [29], [43]. Such change is associated with a preferred direction $\phi = 3\pi/2$. At lower frequencies (i.e., P- or L-band), the scattering return is mainly associated with the trunk structure. In this case, the removal of the tree structure is mapped into a preferred direction $\phi = 3\pi/2$ corresponding to the backscattering decrease in the co-pol channel, and no relevant variation is sensed on the cross-pol one. No relevant scattering changes are sensed from the removal of the canopy. Changes due to the seasonal growth show opposite variations in the canopy and/or the trunk structure. Based on the aforementioned considerations, they are mapped into directions with supplementary angles with respect to those expressed for deforestation change.

A general representation of the multiclass changes that can be detected in dual-pol SAR images by means of the proposed PCV domain is provided in Fig. 1. Fig. 1, which is derived from the above analysis, highlights the PCV properties associated with different kinds of changes and the related multitemporal backscattering behaviors. Note that our analysis covers a wide number of situations, but it cannot be exhaustive. However, following the concepts illustrated here, it is easy to identify the expected position in the PCV domain of other possible kinds of change of interest.

IV. Proposed Technique for CD in the PCV Domain

The framework introduced in Section III provides the main properties of the changed and unchanged pixels in the PCV feature space. The no-change decision region is described by a circle C_{nc} with a radius equal to a threshold value T_ρ, while each of the change classes ω_{ck} can be identified by a sector in the annulus of changed classes A_c, defined by two thresholds T_{k1}, T_{k2}. The interval defined by T_{k1} and T_{k2} includes the preferred direction associated with the change as described in Section III and illustrated in Fig. 1. In real SAR data, the thresholds should be estimated by taking into account the noise effects and, thus, the behavior of the distributions of the magnitude and direction random variables in the PCV feature space. Moreover, without any prior knowledge, the number of change classes in the considered data set is unknown but typically smaller than the aforementioned eight possible preferred directions. Thus, let the expected number of change classes in a specific case be K with $K \in [1, 8]$. In this section, we propose an automatic multiclass CD technique based on the framework presented in Section III. The technique is based on the two steps: 1) a binary CD performed along the magnitude variable, which aims at separating changed from unchanged samples (i.e., classes ω_{nc}, Ω_c) and 2) a multiclass CD performed along the PCV direction variable, which aims at separating different kinds of change $\omega_{ck}, k = 1, \dots, K$ from each other. Fig. 2 shows the block scheme of the proposed automatic and unsupervised technique.

A. Binary CD Along the Magnitude of PCVs

In the first step, the binary CD aims at separating the pixels of the no-change class ω_{nc} from those belonging to the macrochange class Ω_c in an unsupervised and automatic manner. According to the SAR literature, this is typically performed with statistical methods using the distance-based clustering (e.g., K-means [44] or Fuzzy C-means [45]) or thresholding (e.g., Bayesian [4], [12], [46], Otsu [47], and Kittler–Illingworth [11]) decision strategies. Here, we solve the detection problem by defining a strategy for the estimation of statistical distributions for both change and no-change classes along the magnitude variable. The ρ variable carries the information about the presence/absence of change; thus, its probability density function $p_\rho(\rho)$ is expressed as a mixture of two components associated with ω_{nc} and Ω_c

$$p_\rho(\rho) = P(\omega_{nc}) p_\rho(\rho|\omega_{nc}) + P(\Omega_c) p_\rho(\rho|\Omega_c) \qquad (12)$$

where $P(\omega)$ and $p_\rho(\rho|\omega)$ are the prior probability and the marginal distribution for the class ω, $\omega \in \{\omega_{nc}, \Omega_c\}$, respectively. A threshold T_ρ should be employed to separate the two classes. In order to derive T_ρ according to the Bayesian decision strategy, a proper model for the marginal distributions of the mixture needs to be selected. In order to derive a proper model, let us recall that the magnitude is computed from log-ratio elements of PCVs. In the literature, empirical statistical models have been proposed to represent the single-polarimetric log ratio as a mixture of Gaussian or generalized Gaussian distributions [11]. These models show a good trade-off between robustness and estimation accuracy and were used as a preliminary attempt to model the PCV magnitude in [4]. However, the magnitude of a 2-D vector with a zero-mean

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distribution for the vector components has a tailed distribution. Thus, here, we propose to rely on more flexible models like Nakagami [9], Weibull, or generalized Rayleigh [48]. Thus, we consider a mixture model with two Nakagami distributions with priors $P(\omega_{nc})$ and $P(\Omega_c)$, and class-conditional parameters m_{nc}, σ_{nc}, m_c, and σ_c for the no-change and change classes, respectively. The probability density function (pdf) of a single Nakagami distribution with general parameters m (shape) and σ (spread) is defined as follows:

$$p(\rho|\omega) = \frac{2m^m}{\Gamma(m)\sigma^m}\rho^{2m-1}e^{-m\rho^2/\sigma}. \tag{13}$$

Both prior probabilities and marginal unknown parameters of the marginal distributions can be estimated with the EM algorithm according to an interative procedure. The estimates of the prior probabilities and parameters for the marginal distributions at the $(q+1)$th step are obtained by maximizing the estimation of the log-likelihood function at the qth step. The iterative equations for parameter estimation under the assumption of Nakagami mixture model become

$$P^{(q+1)}(\omega) = \frac{\sum_{i=1}^{MN} p^{(q)}(\omega|\rho_i)}{MN} \tag{14}$$

$$\sigma^{(q+1)}(\omega) = \frac{\sum_{i=1}^{MN} p^{(q)}(\omega|\rho_i)\rho_i^2}{\sum_{z=1}^{MN} p^{(q)}(\omega|\rho_z)} \tag{15}$$

$$\sum_{i=1}^{MN} p^{(q)}(\omega|\rho_i)\left[1 - \frac{\rho_i^2}{\sigma^{(q)}} + \log\left(\frac{\rho_i^2}{\sigma^{(q)}}\right) - \psi(m^{(q)}) + \log(m^{(q)})\right] = 0 \tag{16}$$

where $P^{(q)}$, $\sigma^{(q)}$, and $m^{(q)}$ are the prior probability, the spread, and the shape parameters for the class ω at the qth iteration, $\omega \in \{\omega_{nc}, \Omega_c\}$, and $\psi(x) = \Gamma'(x)/\Gamma(x)$ is the digamma function. The iterative process stops when either a maximum number of iterations is reached or the log-likelihood value reaches the convergence condition (i.e., the log-likelihood variation between two consecutive steps is below a tolerance value). The EM algorithm presents a strong sensitivity to the initial conditions, and this does not guarantee the convergence to the global minimum. In order to increase the robustness of the estimation, one of the most popular strategies for the selection of the initial values is represented by the use of the K-means clustering [49]. From the clusters, an initial estimation for the statistical parameters of the two classes is derived. The Bayesian decision rule is finally applied for assigning each sample ρ to either ω_{nc} or Ω_c, that is,

$$\omega^* = \arg\max_{\omega}\{P(\omega)p(\rho|\omega)\}, \omega \in \{\Omega_c, \omega_{nc}\}. \tag{17}$$

This is equivalent to determine the threshold value T_ρ that satisfies the following equation:

$$P(\omega_{nc})p(\rho|\omega_{nc}) = P(\Omega_c)p(\rho|\Omega_c). \tag{18}$$

B. Multiclass CD Along the Direction of PCVs

The second step of the proposed technique consists in the separation of the samples in Ω_c into different change classes $\{\omega_{c1}, \ldots, \omega_{cK}\}$. This is done along the direction variable computed over the distribution of changed pixels $p(\phi|\Omega_c)$ only. $p(\phi|\Omega_c)$ is modeled as a mixture of K components, each associated with a specific change class ω_{ck}, with marginal distributions $p(\phi|\omega_{ck})$ and prior probabilities $P(\omega_{ck}), k = 1, \ldots, K$

$$p(\phi|\Omega_c) = \sum_{k=1}^{K} P(\omega_{ck})p(\phi|\omega_{ck}) \tag{19}$$

where $\sum_{k=1}^{K} P(\omega_{ck}) = 1$. The derivation of the statistical model of the direction feature has been marginally studied in the literature also for passive data. In [22], a statistical nonuniform model has been proposed for the direction feature derived with multispectral data. The model is limited to the case of two nonzero mean Gaussian random variables as components from which deriving the direction and, thus, does not apply to the case of dual-pol SAR images since the X_{LR} components do not satisfy the nonzero mean Gaussian model. In order to introduce a flexible model suitable for the PCV direction variable, we propose to represent the conditional distribution for the kth change class ω_{ck} as a generalized Gaussian function, characterized by three parameters, namely, location μ_k, scale α_k, and shape β_k

$$p(\phi|\omega_{ck}) = \frac{\beta_k}{2\alpha_k\Gamma\left(\frac{1}{\beta_k}\right)}e^{-\left[\frac{|\phi-\mu_k|}{\alpha_k}\right]^{\beta_k}}. \tag{20}$$

An issue to be addressed is that in the multiclass CD, no prior knowledge on the number of changes K is assumed. Here, we automatically estimate its value by minimizing the Bayesian information criterion (BIC) over the discrete set of values $K^* \in [1, 8]$. BIC is one of the state-of-the-art criteria for estimating the quality of model fitting to the data. It takes into account both the complexity of the data model (i.e., the number of free parameters) and the likelihood of the model with respect to the data. By considering (19) and (20), BIC minimization can be written as follows:

$$\begin{aligned} K &= \arg\min_{K^*\in[1,8]} \mathrm{BIC}(K^*) \\ &= \arg\min_{K^*\in[1,8]} (4K^*-1)\log(\mathrm{card}\{\Omega_c\}) - 2\log[\hat{L}(K^*)] \end{aligned} \tag{21}$$

where $\hat{L}(K^*)$ is the maximum log-likelihood obtained by estimating a mixture of K^* components. For each possible K^*, the estimation of the statistical parameters is conducted by applying the following iterative equations for the EM algorithm:

$$L^{(q)}(K^*) = \sum_{z=1}^{\mathrm{card}\{\Omega_c\}} \log\left[\sum_{k=1}^{K^*} P^{(q)}(\omega_{ck})p^{(q)}(\phi_z|\omega_{ck})\right] \tag{22}$$

$$P^{(q+1)}(\omega_{ck}) = \frac{\sum_{z=1}^{card\{\Omega_c\}} p^{(q)}(\omega_{ck}|\phi_z)}{\mathrm{card}\{\Omega_c\}} \tag{23}$$

$$\alpha_k^{(q+1)} = \left[\frac{\beta_k^{(q)}\sum_{z=1}^{\mathrm{card}\{\Omega_c\}} p^{(q)}(\omega_{ck}|\phi_z)\left|\phi_z - \mu_k^{(q)}\right|^{\beta_k^{(q)}}}{\sum_{w=1}^{card\{\Omega_c\}} p^{(q)}(\omega_{ck}|\phi_w)}\right]^{1/\beta_k^{(q)}} \tag{24}$$

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$$\sum_{z=1}^{\text{card}\{\Omega_c\}} p^{(q)}(\omega_{ck}|\phi_z)\frac{\beta_k^{(q)}}{\left(\alpha_k^{(q)}\right)^{\beta_k^{(q+1)}}}\left|\phi_z-\mu_k^{(q+1)}\right|^{\beta_k^{(q)}-1} \cdot \text{sign}\left(\phi_z-\mu_k^{(q+1)}\right)=0 \quad (25)$$

$$\sum_{z=1}^{\text{card}\{\Omega_c\}} p^{(q)}(\omega_{ck}|\phi_z) \times \left[\frac{1}{\beta_k^{(q+1)}}+\frac{\psi\left(\frac{1}{\beta_k^{(q+1)}}\right)}{\left(\beta_k^{(q+1)}\right)^2} - \left(\frac{\left|\phi_z-\mu_k^{(q)}\right|}{\alpha_k^{(q)}}\right)^{\beta_k^{(q+1)}}\log\left(\frac{\left|\phi_z-\mu_k^{(q)}\right|}{\alpha_k^{(q)}}\right)\right]=0. \quad (26)$$

For each K^* value, the estimation process stops when the maximum number of iterations is reached or convergence condition for the likelihood is satisfied. The set of parameters that guarantee the minimum BIC value provides the best fitting solution and the estimated number of changes K. The process is automatic and unsupervised. Starting from the estimated parameters, the multiclass CD map is derived by assigning the class ω_{ck}^* to each considered sample, based on the Bayesian decision rule

$$\omega_{ck}^* = \arg\max_k\{P(\omega_{ck})p(\phi|\omega_{ck})\}. \quad (27)$$

From the posterior probabilities, it is possible to estimate the thresholds T_{k1}^{ϕ}, T_{k2}^{ϕ} for the class ω_{ck}, with $k \leq K$

$$T_{k1}^{\phi}: P(\omega_{ck})p(\phi|\omega_{ck}) = P(\omega_{c(k-1)})p(\phi|\omega_{c(k-1)}) \quad (28)$$

$$T_{k2}^{\phi}: P(\omega_{ck})p(\phi|\omega_{ck}) = P(\omega_{c(k+1)})p(\phi|\omega_{c(k+1)}). \quad (29)$$

In real scenarios, some sectors of the polar domain may not show relevant population (i.e., a given scene does not include all the possible multichange classes). Thus, the thresholds $T_{ki}^{\phi}, k \leq K, i = 1, 2$ will be different from the conceptual theoretical values defined in Section III.

V. Experimental Results

For the validation of the proposed framework, three bitemporal data sets with dual-pol intensity SAR images have been considered, which show different numbers and kinds of change. Two data sets, acquired by the Sentinel-1 and UAVSAR sensors over the areas of Chennai, India, and Los Angeles, CA, USA, respectively, show the presence of two ($K = 2$) relevant changes in the scene. The third data set is more complex. It is acquired by the Sentinel-1 sensor over the area of Oristano, Sardinia, Italy and is characterized by the presence of three changes ($K = 3$). Thus, a total of seven kinds of changes have been analyzed. In order to validate the CD analysis, reference maps have been generated by photo-interpretation, taking the advantage of optical sensors images (i.e., Sentinel-2 data) and ancillary vectorial data. An overview of the considered acquisitions is reported in Table I. Experiments have been performed in MATLAB© environment on a local machine Intel Core i7-4600 CPU 2.10 GHz with 16 GB of RAM. For each data set, PCVs are computed and represented in polar coordinates. CD is conducted in the proposed feature space according to the approach presented in Section IV.

TABLE I
Overview of the Multitemporal Acquisitions for the Experimental Results

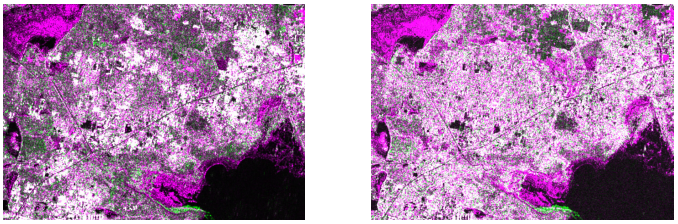
Dataset	Chennai	Los Angeles	Sardinia
Sensor	Sentinel-1	UAVSAR	Sentinel-1
Band	C	L	C
Polarimetric channels	VV-VH	HH-HV	VV-VH
Resolution	$20 \times 20m$	$3 \times 3m$	$20 \times 20m$
Acquisitions	Oct. 31, 2015 Nov. 12, 2015	Apr. 23, 2009 May 11, 2015	May 21, 2016 July 20, 2016

Two thresholding methods have been tested for both the PCV magnitude and direction: Bayesian thresholding with the parameters estimated by the EM algorithm and Otsu thresholding. The optimal number of change classes K was automatically determined by BIC considering the analysis of a mixture of K^* generalized Gaussians with K^* integer ranging in $[1, 8]$. CD results obtained by the proposed framework and employing the statistical models illustrated in Section IV were compared with: 1) the preliminary version of the proposed method in [4] that uses simpler models for both the magnitude variable (i.e., generalized Gaussian mixture) and the direction (i.e., Gaussian mixture) (we refer to this method as simplified polar analysis, SPA); 2) a binary CD method defined for PolSAR data based on the log-likelihood ratio (LLR) and on the use of a Gamma statistical model for each scattering amplitude [17]; and 3) a multiclass CD method based on a clustering approach that considers PolSAR multitemporal information as a mixture of bivariate Gaussian components (EM2DGM) but not designed for PolSAR data [21]. Unknown parameters of the mixture are estimated by the EM algorithm. The two reference multiclass methods require the prior knowledge of the number of classes in the mixture. This information is injected by using the K value obtained by the proposed automatic approach. This favors the reference methods with respect to the proposed one that estimates K in an unsupervised and automatic way. Accuracy assessment is performed in terms of two popular indices, namely, overall accuracy (OA) and Kappa coefficient (Kappa) [50], [51]. The former provides an overall measure of the correct classification without taking into account the prior probability of different classes. The latter takes into account the class priors for weighting errors.

A. Chennai Data Set

The Chennai data set includes two dual-pol SAR GRD-mode images acquired by the Sentinel-1 SAR sensor. Both images have a spatial resolution of 20×20 m and data in VV co-pol and VH cross-pol channels. They have been acquired over the area of Chennai in October 3, 2015 and November 12, 2015, respectively. During that period, the area has been hit by a flood, which interested the rural areas of the region and, in minor part, the suburbs of the city. In order to proceed with the analysis, the two SAR images

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(a) (b)

Fig. 3. Chennai data set: multitemporal backscattering false-color composition for (a) VV channel and (b) VH channel. R, B: October 2015 and G: November 2015. Backscattering increase and decrease are represented in green and magenta, respectively.

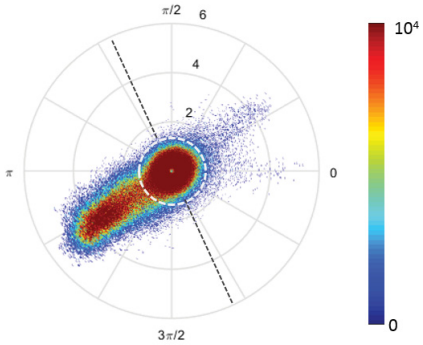


Fig. 4. Chennai data set: magnitude mixture model. Chennai data set: bivariate polar histogram of the PCVs. The estimated $T_\rho = 1.3$ dB (white dashed circle) and $T_{12} = -1.2$ rd (black dashed line) are shown.

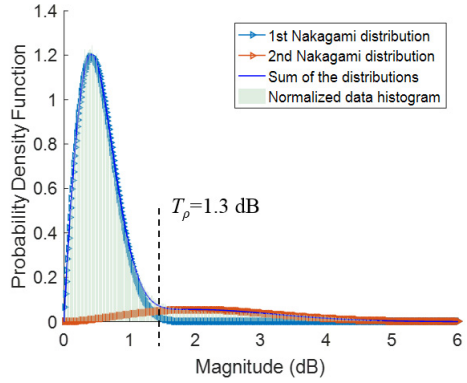
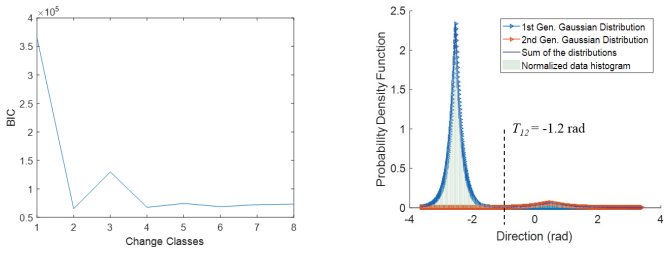


Fig. 5. Chennai data set: magnitude mixture model.



(a) (b)

Fig. 6. Chennai data set. (a) BIC value versus K value (minimum value is reached for $K = 2$). (b) Direction histogram and estimated mixture model for $K = 2$.

have been preprocessed by applying radiometric calibration, co-registration, and noise filtering [16].

A portion of 686×963 pixels has been considered for a quantitative analysis. Fig. 3(a) and (b) shows the multitemporal false-color composition of the backscattering for the VV and VH polarimetric channels, respectively. Gray shades correspond to no change, whereas the increase and decrease of backscattering appear in green and magenta, respectively. A reference map has been generated starting from ground truth vectorial data from the United Nations Institute for Training And Research (UNITAR), Geneva, Switzerland [52]. The reference map represents the changes of the water bodies after the flood event [see Fig. 7(a)] only. PCV with $X_{\mathrm{LR}}^{\mathrm{VV}}$ and $X_{\mathrm{LR}}^{\mathrm{VH}}$ components has been generated for each pixel. By representing the PCV in the polar coordinate system (see Fig. 4), we can see that the flooded pixels tend to cluster in a sector with a preferred direction $\phi = 5\pi/4$, as the presence of a new water body is associated with a sensible decrease of $X_{\mathrm{LR}}^{\mathrm{VV}}$. Regions where the water body disappeared tend to cluster along a preferred direction $\phi = \pi/4$ in agreement with the model in Section III.

By applying the proposed CD approach, the EM estimation of the priors and marginal distributions of the classes ω_{nc} and Ω_c results in the PCV magnitude distribution illustrated in Fig. 4. The obtained Bayesian threshold for separating ω_{nc} and Ω_c is $T_\rho = 1.3$ dB. In order to estimate the optimal number of change classes, the PCV direction distribution has been modeled as a mixture of K generalized Gaussian distributions. Fig. 6(a) shows the plot of the BIC criterion with respect to the number of classes. The smallest BIC indicates an estimated value $K = 2$. The result is qualitatively confirmed by the trend of the direction pdf. Accordingly, the EM algorithm has been applied fixing $K = 2$ for the estimation of the priors and the marginal distribution parameters. Threshold values $T_{12} = T_{21} = -1.2$ rd have been estimated with the Bayesian decision rule. Fig. 6(b) shows the direction normalized histogram and the estimated class distributions. Fig. 7(b) shows the multiple CD map.

By analyzing Fig. 7(a) and (b), we can observe that the proposed approach obtained reliable results with the correct detection of both the flooded areas around the lake and the strip of land where the water disappeared. A small amount of errors is related to false alarms occurring on the change class ω_{c2} with preferred direction $\phi = \pi/4$. These errors are associated with backscattering variations located on small urban areas [purple circle in Fig. 7(b)] probably due to the rain causing the flood. A large amount of errors (in terms of both false and missed alarms) is associated with the change class ω_{c1} with preferred direction $\phi = 5\pi/4$. False alarms [orange circle in Fig. 7(b)] are due to soil backscattering variations, which may be due to temporal variations of the surface roughness parameters, not reported in the water-body reference map. On the other hand, missed detections for the class ω_{c2} [green circle in Fig. 7(a)] may be partially explained by a lower level of accuracy of the water-body boundaries in the preevent map, which does not show the presence of all the small water regions. The entire CD process took 1099 s.

The quantitative analysis points out that the use of the proposed framework with the novel statistical models and both Bayesian and Otsu thresholding strategies achieves good performance in terms of both OA (i.e., 0.9307 and 0.9432) and Kappa (i.e., 0.6668 and 0.6910) (see Tables II and III). The

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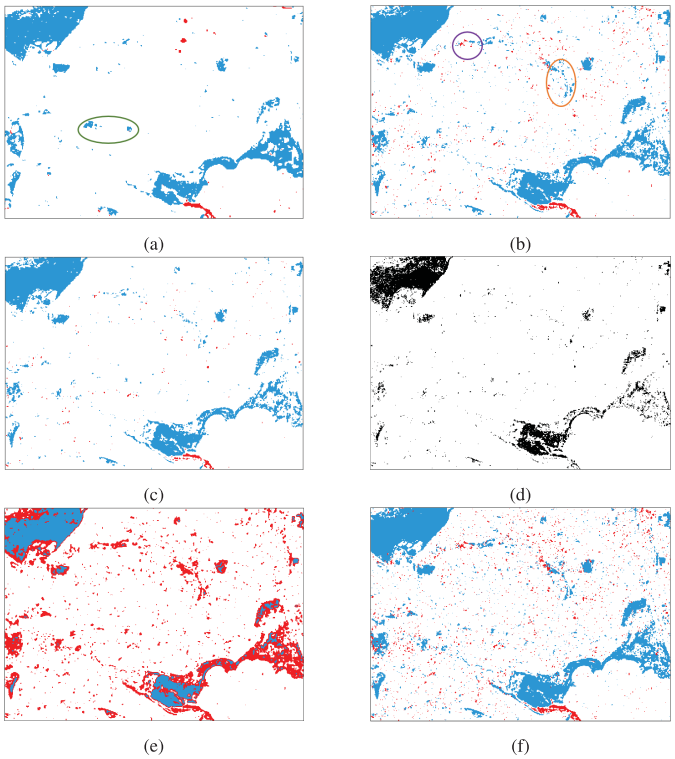


Fig. 7. Chennai data set maps. (a) Reference. (b) Proposed (Bayesian thresholding). (c) Proposed (Otsu thresholding). (d) LLR. (e) EM2DGM. (f) SPA (ω_{nc}—white, Ω_c—black, ω_{c1}—blue, and ω_{c2}—red).

TABLE II

CHENNAI DATA SET: CONFUSION MATRIX AND OA FOR THE PROPOSED APPROACH

	$\hat{\omega}_{nc}$	$\hat{\omega}_{c1}$	$\hat{\omega}_{c2}$
ω_{nc}	560573	24356	5274
ω_{c1}	14693	53366	68
ω_{c2}	1363	38	887
		OA	0.9307

Otsu thresholding yields slightly higher performance that can be explained by the pronounced separation of the modes in the histograms of both magnitude and direction on this data set. The use of the simplified models in [4] provided lower performance: OA reduced by 4% (i.e., 0.9114) and Kappa reduced by 8% (i.e., 0.6156) because of the lower accuracy in fitting capabilities [see Fig. 7(f)]. Despite the proposed approach showed OA similar to that of the LLR approach, it slightly increased the Kappa accuracy of about 14%. This is because the proposed framework distinguishes among different kinds of change [see Fig. 7(b)–(d)], whereas LLR does not and this impacts on Kappa. The improvement with respect to the EM2DGM is of about 20%, even if an advantage is given to the ED2DGM by providing as input the number of change classes, estimated automatically by the proposed approach. Furthermore, the proposed approach shows also a better discrimination between the two change classes together with a discrimination improvement for the change/no change. Table III summarizes the results.

B. Los Angeles Data Set

The Los Angeles data set considers two full-pol SAR images acquired by the UAVSAR airborne sensor on the Los

TABLE III

CHENNAI DATA SET: PERFORMANCE COMPARISON WITH THE STATE-OF-THE-ART METHODS

Method	Multiple CD				Binary CD
	Proposed		EM2DGM	SPA	LLR
	Bayesian	Otsu	[**?**]	[4]	[17]
OA	0.9307	0.9432	0.8749	0.9114	0.9305
Kappa	0.6668	0.6910	0.4907	0.6156	0.5255

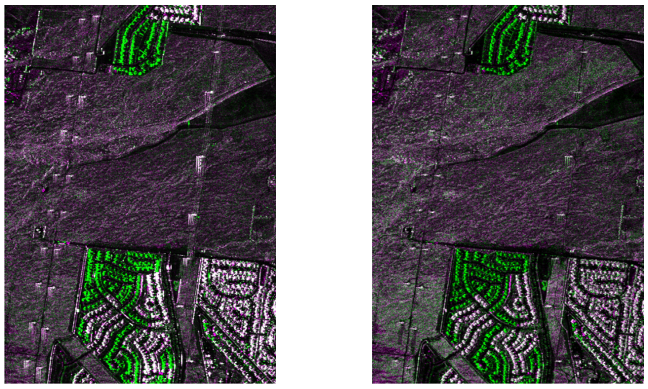


Fig. 8. Los Angeles data set: multitemporal backscattering false-color composition for (a) HH channel and (b) HV channel. R, B: April 2009 and G: May 2015. Backscattering increase and decrease are represented in green and magenta, respectively.

Angeles area. The data are full-polarimetric with a spatial resolution of 0.4×1.6 m. For our analysis, intensity information from the two polarimetric channels HH and HV was considered. The two images have been acquired on April 23, 2009 and May 11, 2015, respectively. The area has been interested by urbanization with the construction of new buildings and the vegetation clear-cut for future settlements. The two PolSAR images have been preprocessed by applying radiometric calibration, co-registration, and multilooking for speckle noise mitigation. The multilooked images are characterized by a spatial resolution of 3×3 m. A crop of 559×400 pixels has been considered. Fig. 8(a) and (b) shows the multitemporal false-color composition of the backscattering for the HH and HV polarimetric channels, respectively. Due to the lack of ancillary data for the proposed scene, a reference map has been generated by photo-interpretation [see Fig. 12(a)] from the Pauli RGB false-color composition of the two images by visual inspection with the support of ancillary optical data (i.e., Landsat 5/Landsat 7).

The multitemporal comparison has been applied to the polarimetric images to compute the PCVs. Fig. 9 shows the bivariate data distribution in the PCV polar coordinate system. The new built-up areas are characterized by the positive values for both the PCV components. According to the analysis in Section III, the change locates in a sector of the feature space with preferred direction $\phi = \pi/4$. On the other hand, pixels associated with a vegetation clear-cut have negative values for $X_{\mathrm{LR}}^{\mathrm{HV}}$. This corresponds to a sector with preferred direction $\phi = 5\pi/4$ in the PCV features space as expected from our theoretical analysis.

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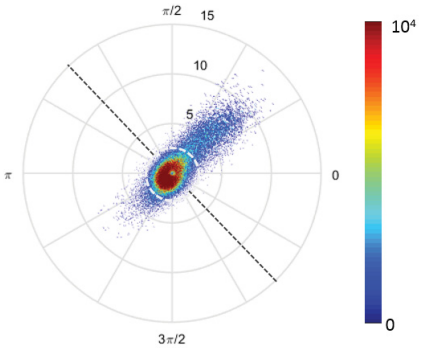


Fig. 9. Los Angeles data set: bivariate polar histogram of the PCVs. The estimated $T_\rho = 2.4$ dB (white dashed circle) and $T_{12} = -0.85$ rd (black dashed line) are shown.

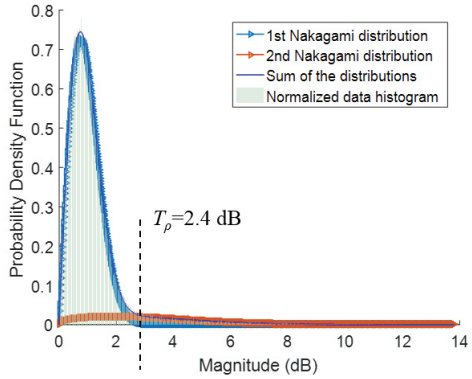


Fig. 10. Los Angeles data set: magnitude mixture model.

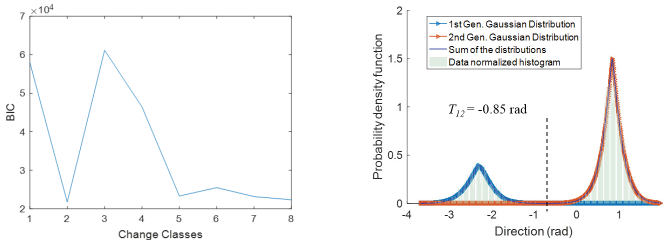


Fig. 11. Los Angeles data set. (a) BIC analysis with respect K (minimum value is reached for $K = 2$). (b) Direction histogram and estimated mixture model for $K = 2$.

After binary CD on the magnitude feature (see Fig. 10), the multiclass CD has been applied to changed pixels only. Fig. 11(a) shows the behavior of BIC versus the estimated number of classes. The smallest BIC value is associated with $K = 2$. The EM algorithm has been applied with $K = 2$ for the estimation of the priors and marginal distribution parameters. Threshold values $T_{12} = T_{21} = -0.85$ rd have been estimated by the Bayesian decision rule. Figs. 11(b) and 12(b) show the direction pdf of the change classes and the multiclass CD map. The computational time required for this data set was 388 s.

The proposed approach shows OA = 0.9671 and Kappa = 0.6961 when employing the Bayesian decision theory (see Table IV). Higher performance (i.e., an increase of 2% for Kappa) is yielded by considering the Otsu thresholding, favored by the pronounced mode separation. The use of less accurate statistical models as in [4] reduces the OA of about 2% (i.e., 0.9544) and the Kappa of about 8% (i.e., 0.6340) [see Fig. 12(f)].

The proposed framework for both cases shows OA improvement of about 4% compared to the one of EM2DGM, with

TABLE IV

LOS ANGELES DATA SET: CONFUSION MATRIX AND OA FOR THE PROPOSED APPROACH

	$\hat{\omega}_{nc}$	$\hat{\omega}_{c1}$	$\hat{\omega}_{c2}$
ω_{nc}	207241	2755	2489
ω_{c1}	383	1014	0
ω_{c2}	1721	0	7997
		OA	0.9671

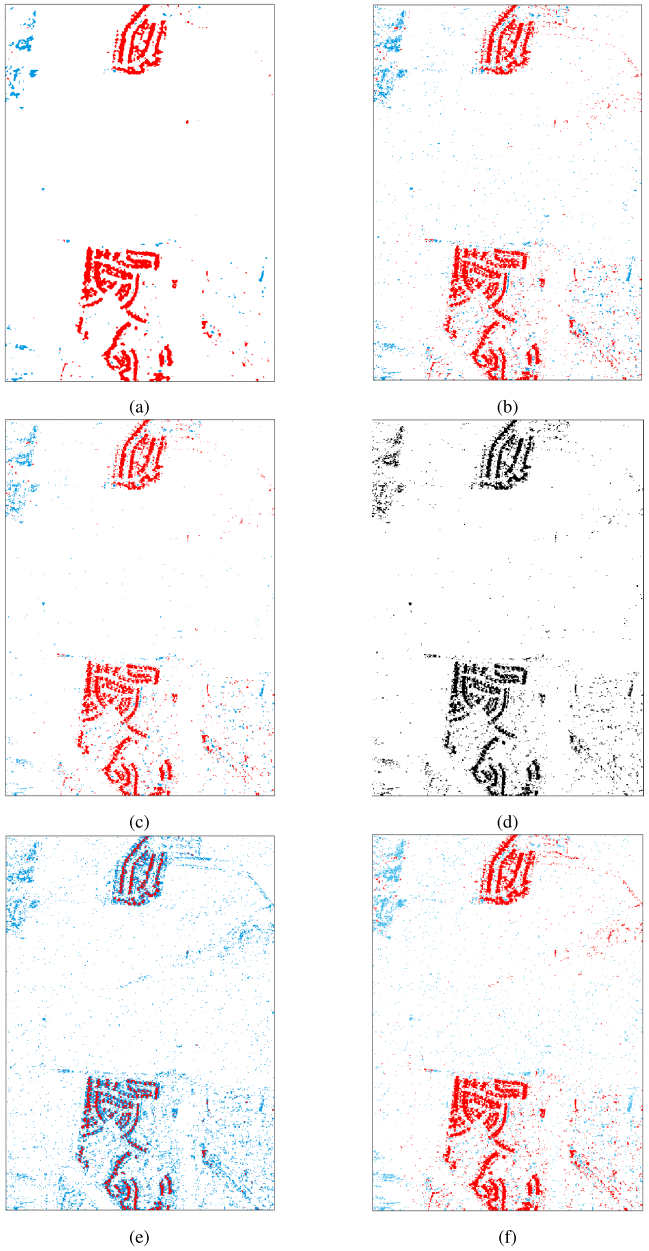
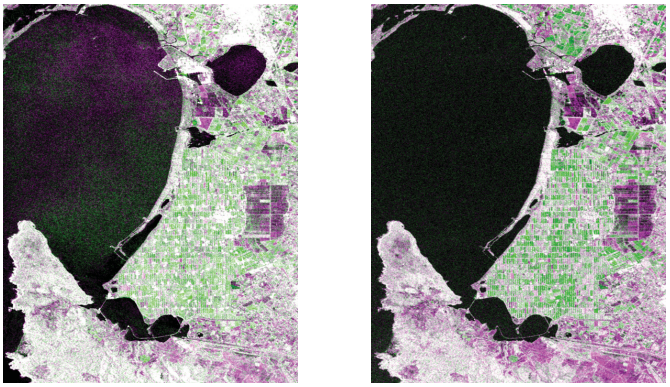


Fig. 12. Los Angeles data set maps. (a) Reference. (b) Proposed (Bayesian thresholding). (c) Proposed (Otsu thresholding). (d) LLR. (e) EM2DGM. (f) SPA (ω_{nc}—white, Ω_c—black, ω_{c1}—blue, and ω_{c2}—red).

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TABLE V
LOS ANGELES DATA SET: PERFORMANCE COMPARISON WITH THE STATE-OF-THE-ART METHODS

Method	Multiple CD				Binary CD
	Proposed		EM2DGM [?]	SPA [4]	LLR [17]
	Bayesian	Otsu			
OA	0.9671	0.9734	0.9257	0.9544	0.9733
Kappa	0.6961	0.7218	0.4546	0.6340	0.7178



(a) (b)

Fig. 13. Sardinia data set: multitemporal backscattering false-color composition for (a) HH channel and (b) HV channel. R, B: May 2016 and G: July 2016. Backscattering increase and decrease are represented in green and magenta, respectively.

value similar to the ones of LLR. However, the LLR final result [see Fig. 12(d)] does not distinguish among different kinds of change. Despite the advantage given to the ED2DGM by providing as input the number of change classes estimated by the proposed approach, the proposed method improved Kappa of more than 24% with respect to EM2DGM. Furthermore, the proposed approach shows also a better discrimination between the two change classes together with a discrimination improvement for the change/no change. A comparison between Fig. 12(b) and (e) and the reference map in Fig. 12(a) points out that the proposed method better identifies new built-up areas (red) than the EM2DGM. Table V summarizes the results.

C. Sardinia Data Set

The Sardinia data set includes two dual-pol SAR GRD images acquired by the Sentinel-1 SAR sensor with a spatial resolution of 20 × 20 m and two polarimetric channels, namely, VV and VH. The two images have a size of 900 × 700 pixels and have been acquired on May 21, 2016 and July 20, 2016 over the area of Oristano. The area has been interested by multiple changes. A relevant change of the surface roughness interested local areas of the sea surface and water bodies with a decrease of the swell [53]. Other changes relate to the agricultural fields. Based on the proposed scattering model, a class of change includes the increase of dense vegetation and tillage on bare soils, respectively. The other change class includes opposite changes, due to the vegetation decrease, associated with seasonal harvesting and small fires. From a backscattering perspective, the change on the sea surface is modeled as a variation on the co-polarimetric

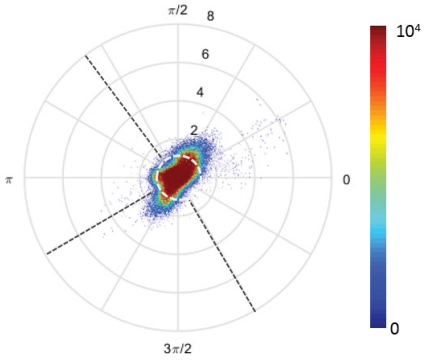


Fig. 14. Sardinia data set: bivariate polar histogram of PCVs. Estimated $T_\rho = 0.9\ dB$ (white dashed circle), $T_{12} = 2.24$ rd, and $T_{23} = 3.55$ rd (black dashed lines) are shown.

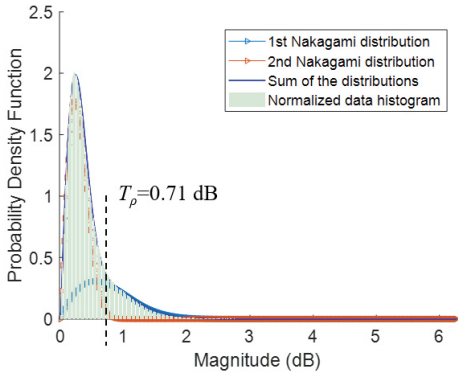


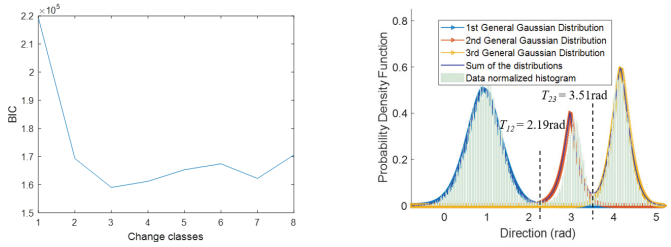
Fig. 15. Sardinia data set: magnitude mixture model.

channel only, while the changes on the vegetation are modeled by backscattering variations on both the channels, as illustrated in Section III. This data set shows a larger complexity compared to the two previous ones, due to the large size, the larger number of changes, and their backscattering complexity. The two PolSAR images have been preprocessed by applying radiometric calibration, co-registration, and filtering. Fig. 13(a) and (b) shows the multitemporal false-color composition of the backscattering for the VV and VH polarimetric channels, respectively.

A reference map has been generated by photo-interpretation from a pair of Sentinel-2 optical images acquired on the same region on May 22, 2016 and July 18, 2016, respectively. A visual inspection was conducted on the co-polarimetric channel data for tracing water regions with changes in the surface roughness. Fig. 17(a) shows the reference map.

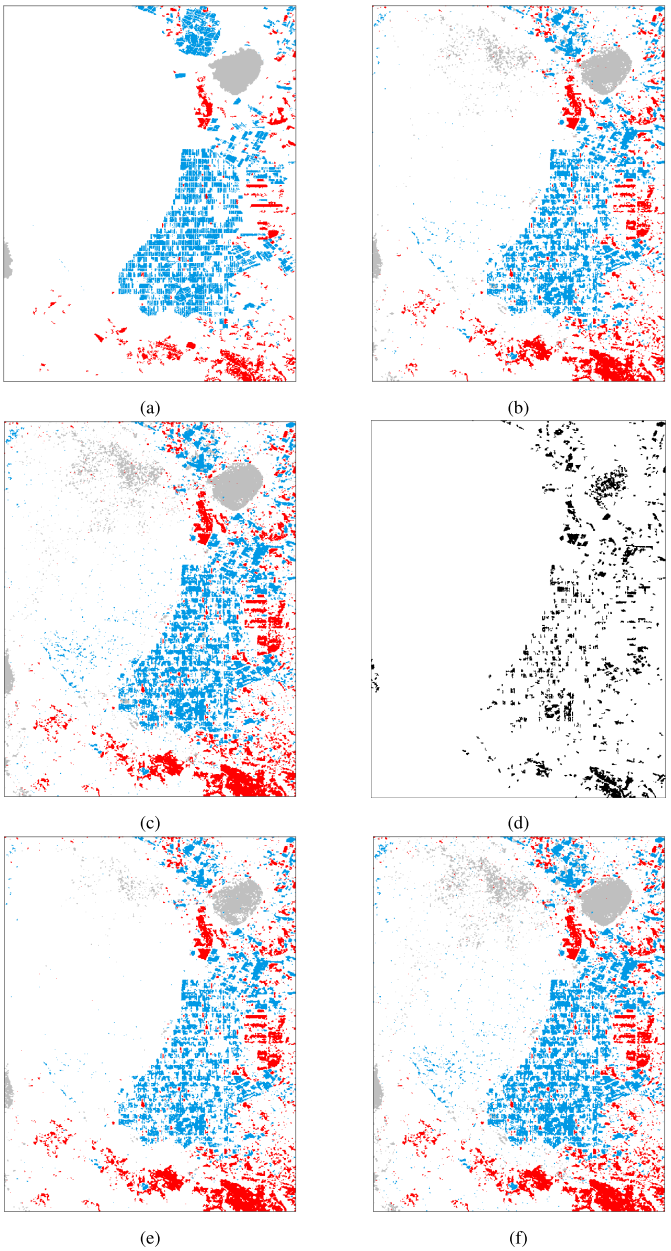
A PCV with $X_{\mathrm{LR}}^{\mathrm{VV}}$ and $X_{\mathrm{LR}}^{\mathrm{VH}}$ components has been derived for each pixel. By representing the PCVs in the polar coordinate system (see Fig. 14), we can see that the pixels affected by a decrease of vegetation tend to cluster in a sector with a preferred direction $\phi = 5\pi/4$, associated with negative values for both $X_{\mathrm{LR}}^{\mathrm{VH}}$ and $X_{\mathrm{LR}}^{\mathrm{VV}}$. The opposite log-ratio values are expected for pixels associated with fields growth with a corresponding preferred direction $\phi = \pi/4$. Pixels affected by a decrease of the sea swell are expected to fall in the sector with preferred direction $\phi = \pi$ in agreement with the model in Section III.

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(a) (b)

Fig. 16. Sardinia data set. (a) BIC analysis with respect K (minimum value is reached for $K = 3$). (b) Direction histogram and estimated mixture model for $K = 3$.



(a) (b) (c) (d) (e) (f)

Fig. 17. Sardinia data set maps. (a) Reference. (b) Proposed (Bayesian thresholding). (c) Proposed (Otsu thresholding). (d) LLR. (e) EM2DGM. (f) SPA (ω_{nc}—white, Ω_c—black, ω_{c1}—blue, ω_{c3}—gray, and ω_{c3}—red).

The proposed multiclass CD approach has also been applied to this data set. Fig. 15 shows the estimated distributions for the two classes. Bayesian thresholding has been applied for separating them. Outliers with a size of one pixel due to residual noise were removed. The multiclass CD has been applied to the set of changed pixels only. Fig. 16(a) shows the plot of the BIC value versus the number of classes. The minimum BIC value is obtained for $K = 3$. The EM algorithm has been applied with $K = 3$ to the estimation of the priors and the marginal distribution parameters. Fig. 16(b) shows the direction pdf and the estimated class distributions. The entire process took 1149 s. Fig. 17(b) shows the final CD map in the case of Bayesian thresholding. The quantitative performance analysis resulted in OA = 0.8961 and Kappa = 0.6084 (see Table VI). For this more complex scene, with a more critical separation between the change and no-change classes along the magnitude with respect to the previous experiments, the Bayesian thresholding is more accurate than the Otsu one. Also in this case, the use of less accurate models as SPA reduces the performance in terms of Kappa of about 5% (i.e., 0.9669) and OA of about 3% (i.e., 0.5609).

The proposed framework allows for OA values similar to those of LLR [see Fig. 17(d)] and EM2DGM [see Fig. 17(e)]. However, Kappa has an improvement of about 18% and 2% with respect to LLR and EM2DGM, respectively. Also in this case, LLR provides a binary map only. Table VII summarizes the results on this data set.

TABLE VI

SARDINIA DATA SET: CONFUSION MATRIX AND OA FOR THE PROPOSED APPROACH

	$\hat{\omega}_{nc}$	$\hat{\omega}_{c1}$	$\hat{\omega}_{c2}$	$\hat{\omega}_{c3}$
ω_{nc}	504814	18494	7117	15892
ω_{c1}	20352	33593	993	99
ω_{c2}	551	14	10156	25
ω_{c3}	1285	561	90	15964
			OA	0.8961

TABLE VII

SARDINIA DATA SET: PERFORMANCE COMPARISON WITH THE STATE-OF-THE-ART METHODS

Method	Multiple CD				Binary CD
	Proposed		EM2DGM	SPA	LLR
	Bayesian	Otsu	[?]	[4]	[17]
OA	0.8961	0.8714	0.8918	0.8669	0.9026
Kappa	0.6084	0.5669	0.5849	0.5609	0.4307

VI. CONCLUSION AND DISCUSSION

In this article, a novel framework for the analysis of multiclass change information in multitemporal dual-pol SAR data has been proposed. The framework relies on the backscattering information in two polarimetric channels, namely, a co-polarized and a cross-polarized and on the novel concept of PCV feature space for representing a 2-D vector with the polarimetric multitemporal change information. The backscattering information in the PCV feature space makes the no-change and change classes representable in polar coordinates as fitting into a circle and an annulus, respectively. In particular, physical changes occurring in the scene can be mapped in specific sectors in the polar domain according to

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their backscattering variations in polarimetric channels. Such mapping is based on the theoretical/conceptual analysis of the single-date backscattering behaviors and sensitivity to target main parameters (i.e., surface roughness, dielectric constant, and the presence of dihedral structures) and their expected multitemporal variation in the case of changes. Based on the proposed framework, a two-step multiclass CD approach has been derived, which separates the no-change and multiclass changes in the PCV feature space after automatically detecting the number of changes. The approach relies on a novel statistical model for probability density functions in the PCV domain.

Experimental results proved the reliability of the proposed framework and the effectiveness of the CD technique on three data sets showing seven different multiclass changes. A comparison with multidimensional clustering approaches pointed out that the proposed approach has both a better discrimination capability of the no-change class (by means of the problem decomposition along the magnitude variable) and a better multiclass CD performance (because of the explicit use of the scattering information of the physical target properties for modeling multitemporal changes in the direction variable).

Future developments will investigate the multitemporal backscattering information for all the polarimetric channels in order to extend the proposed model to full-polarimetric data and thus mitigating the limitations of dual-pol SAR images. In addition, the use of complex PolSAR information in the multitemporal analysis for the generation of multiclass CD maps will be studied to enhance the information yielded by backscattering intensity on the polarimetric channels.

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